SPE-227153-MS

# Study on Fluid Potential Calculation Method and Adjustment Strategy of Fractured-Caved Carbonate Reservoirs

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.



## Abstract

Fractured-caved carbonate reservoirs typically exhibit multi-medium characteristics, comprising pores, fractures, and large-scale cavities. The coexistence of these complex flow systems gives rise to intricate fluid migration behavior and presents considerable challenges in controlling water production and enhancing oil recovery. Adjusting the fluid potential based on reservoir hydrodynamics is considered an essential approach for well pattern management in such reservoirs. Nevertheless, the existing fluid potential regulation theories are still constrained by incomplete formulations and a lack of reliable quantitative characterization tailored for fractured-caved carbonate settings. To address these limitations, this study develops a fluid potential calculation and characterization methodology specifically adapted to the flow mechanisms of fractured-caved carbonate reservoirs. Through a comparative analysis of the relative magnitudes of various potential energy components within fractured-caved units, the primary controlling factors influencing the fluid potential distribution were systematically identified, leading to the establishment of a theoretical representation model. Based on this framework, bottom-hole pressure data from representative production wells were processed to normalize the fluid potential distribution across the reservoir units. By incorporating production performance analysis, development potential assessment, and inter-well connectivity evaluation, well-specific adjustment strategies were formulated, and a practical fluid potential adjustment policy chart was constructed. The proposed methodology was applied to 27 well groups in the T10 oilfield of the Tarim Basin, China. The calculated fluid potential fields for different fractured-caved units enabled the design of differentiated adjustment schemes targeting specific potential wells. Field implementation over a five-year period resulted in a cumulative oil production increase of approximately 746000 t, while maintaining the average water cut consistently between 75% and 85%. These outcomes demonstrate the reliability and practical value of the proposed approach. The fluid potential calculation method and the adjustment policy chart presented in this work offer effective technical solutions for optimizing the development of similar fractured-caved carbonate reservoirs.

**Keywords:** fractured-caved carbonate reservoirs; fluid potential; development potential evaluation; reservoir production optimization; hydrodynamic adjustment strategy

## Introduction

Carbonate reservoirs are widely found in regions such as the Middle East, North America, South America, and Asia. These reservoirs hold about 52% of the world's proven recoverable reserves and contribute nearly 60% of global oil and gas production (Gao et al., 2025; Li et al., 2018; Tong et al., 2024; Zhou et al., 2024). In China, fractured-caved carbonate reservoirs are mainly located in the Tarim Basin and the Bohai Bay Basin, and they play an important role in increasing oil reserves and boosting production (Li, 2013). The Ordovician carbonate reservoirs in the Tarim Basin have gone through multiple tectonic uplifts and changes in sea level. Long-term karst processes during the Caledonian and Hercynian periods have formed complex fracture-cave systems (Jiang et al., 2024; Zhang et al., 2021a, 2023). Based on reservoir space types and their contributions to oil production, fractured-caved carbonate reservoirs can be divided into four main types: cave-dominated, vuggy, fracture-dominated, and matrix-dominated reservoirs. These reservoirs usually have poor connectivity and strong heterogeneity (Hu et al., 2023; Zhang et al., 2021b). The oil-water distribution in these reservoirs is often complex and is strongly affected by local fracture-cave units. Sealed water bodies and strong bottom aquifers are common. These features can lead to shorter water-free production periods, rapid water cut increase after breakthrough, and serious difficulties in water control, which greatly limit oil recovery (Cheng et al., 2023; Zhao et al., 2023; Zheng et al., 2023). In some wells, the bottom aquifer is not connected, which can cause low reservoir energy in the later production stage (Tang et al., 2023). As a result, keeping oil production stable while controlling water cut has become a key challenge in the development and adjustment of fractured-caved carbonate reservoirs in the Tarim Basin.

At present, numerical reservoir simulation has become a widely used method for developing and adjusting fractured-caved carbonate reservoirs. This approach is frequently applied in well pattern optimization, water control, production improvement, and low-efficiency well group management. Li et al. (2020) investigated the low efficiency of water flooding in fractured-caved carbonate reservoirs through numerical simulation. Their study analyzed the relationship between fracture-cave structures and water flooding performance under different karst conditions and proposed targeted injection-production strategies. Cheng (2022) performed numerical simulation of hydrocarbon gas injection for enhanced oil recovery in fractured-caved carbonate reservoirs. Based on geological modeling, history matching, remaining oil characterization, and phase behavior analysis, the study clarified the minimum miscibility conditions and optimized the injection scheme. Li et al. (2025) established numerical simulation models for different karst development patterns by considering the vertical fluid equilibrium effect. They quantified the adjustment schemes and injection-production parameters based on remaining oil distribution and revealed the influence of karst patterns on production characteristics, remaining oil occurrence, and development strategies. However, most of these simulation studies are based on the continuum medium theory, which cannot accurately represent the discontinuous and irregular geometries between fractures and caves. To address this issue, Zheng et al. developed a discrete model for fractured-caved carbonate reservoirs, which significantly improved the description of fracture-cave structures and effectively overcame the limitations of continuum models in simulating water cut evolution and multiphase flow behavior (Wang et al., 2024; Zhang et al., 2022; Zheng et al., 2024a, 2024b). Nevertheless, numerical simulation methods still heavily rely on detailed geological models, which require substantial workload and long modeling cycles, making it difficult to achieve efficient, real-time dynamic management of low-efficiency well groups in the field.

In recent years, reservoir adjustment techniques based on fluid potential analysis have received growing attention in the development of fractured-caved carbonate reservoirs. The fluid potential field is closely associated with hydrocarbon migration, accumulation, and trap formation. Typically, oil and gas migrate along the fluid potential gradient from high-potential areas to low-potential zones, where accumulation tends to occur (Cheng et al., 2016; Pang et al., 2012). Fluid potential integrates the driving effects of kinetic energy, pressure, buoyancy, and capillary forces, providing a comprehensive description of subsurface fluid movement. By analyzing the spatial distribution of the fluid potential field, it is possible to diagnose production conflicts and design effective reservoir adjustment strategies. Ren (2019) proposed a water control and production improvement strategy by regulating the production pressure differential to adjust the fluid potential in fractured-caved carbonate reservoirs. This study demonstrated that fluid potential analysis has significant potential in stabilizing oil production and controlling water cut. Hou et al. (2020) further examined the influence of multiple factors on fluid potential adjustment and established the relationship between the liquid production ratio and the adjustment effectiveness under optimal water-oil volume ratio conditions. Du et al. (2020) explored the evolution of fluid potential and remaining oil distribution based on four representative fracture-cave structures and identified the key factors controlling potential regulation. Li et al. (2021) developed a fluid potential theoretical model for fractured-caved carbonate reservoirs by considering different reservoir types and development stages. The study systematically clarified the mathematical calculation methods for various potential components and defined the application boundaries of fluid potential regulation techniques. Compared with traditional numerical simulation approaches, fluid potential analysis does not rely on geological modeling and offers advantages such as simplified calculations, fast processing, and satisfactory accuracy. However, due to the complex characteristics of fractured-caved carbonate reservoirs, the existing fluid potential characterization methods remain incomplete. The calculation processes involve multiple parameters, and the workflows are often complex. Furthermore, related adjustment techniques are still under exploration, and a systematic theoretical framework has not yet been fully established, which limits their ability to provide reliable, practical guidance for field development.

To address these challenges, this study presents a fluid potential calculation method and a corresponding adjustment strategy specifically designed for fractured-caved carbonate reservoirs. The T10 oilfield in the Tarim Basin, China, with 27 well groups, was selected as the case study. Based on the traditional fluid potential theory, a comparative analysis of the orders of magnitude of different potential components within fractured-caved development units was conducted to identify the primary factors controlling the fluid potential in the hydrodynamic system of these reservoirs. On this basis, a theoretical model for fluid potential characterization in fractured-caved carbonate reservoirs was developed. Field bottom-hole pressure data from production wells in typical fractured-caved units were processed to obtain the distribution of the fluid potential field. By integrating production performance analysis, development potential evaluation, and inter-well connectivity assessment, well-specific fluid potential adjustment strategies were established for each typical fractured-caved unit. Furthermore, a fluid potential adjustment policy chart was constructed to define the regulation boundaries for different types of fractured-caved units, providing technical guidance for the efficient development of similar reservoirs.

## Fluid Potential Characterization and Calculation

### Fluid Potential Characterization

The concept of fluid potential was first introduced by M.K. Hubbert in the 1940s and 1950s, who defined it as the mechanical energy per unit mass of fluid. Fluid potential is primarily composed of gravitational potential energy, elastic potential energy, and kinetic energy (Hubbert, 1940, 1953). In 1987, W.A. England further refined Hubbert's theory by incorporating the effects of interfacial forces and capillary pressure into the fluid potential framework (England et al., 1987). Since then, fluid potential analysis has been widely applied in oil and gas field development to evaluate the driving forces of hydrocarbon migration. In this context, the analysis generally considers gravitational potential energy ($E_g$), pressure potential energy ($E_p$), kinetic energy ($E_k$), and interfacial energy ($E_\sigma$), which respectively correspond to gravitational forces, formation pressure, hydrodynamic forces, viscous forces, and capillary effects (Ren, 2019). According to fluid potential theory, the total potential energy of a unit volume of fluid can be expressed as follows:

$$\Phi = \rho gz + \rho\int_0^p \frac{dp}{\rho(p)} + \rho\frac{v2}{2} + \frac{2\sigma\cos\theta}{r} \tag{1}$$

where the gravitational potential is defined as $\Phi_1 = \rho gz$, the pressure potential as $\Phi_2 = \rho\int_0^p \frac{dp}{\rho(p)}$, the kinetic potential as $\Phi_3 = \rho\frac{v2}{2}$, and the interfacial potential as $\Phi_4 = \frac{2\sigma\cos\theta}{r}$.

Based on the above definitions, the magnitudes of each potential component in the T10 reservoir were calculated (Table 1). The results indicate that, in comparison to gravitational and pressure potentials, both the kinetic and interfacial potentials are orders of magnitude smaller. Their contributions to the total potential are minimal and can typically be neglected. As a result, kinetic and interfacial potentials are usually excluded from practical fluid potential analysis.

**Table 1—The fluid potential in the T10 reservoir is mainly composed of gravitational, pressure, kinetic, and interfacial components.**

| Gravitational Potential (MPa) | Pressure Potential (MPa) | Kinetic Potential (MPa) | Interfacial Potential (MPa) |
|---|---|---|---|
| < 3.5 | > 50 | $< 10^{-4}$ | $< 10^{-7}$ |

Gravitational potential refers to the gravitational energy of subsurface fluids relative to a reference plane. In this study, the T74 interface was selected as the reference plane. The relative elevation was determined by calculating the difference between the well depth and the top depth of the Ordovician formation (see Figure 1). Fluid density was obtained from well logging, core sampling, and well testing data.

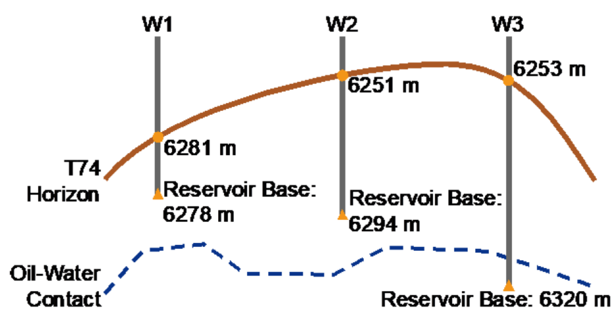


**Figure 1—Calculation of the vertical distance between the bottom boundary of the pay zone and the T74 reference surface. The relative elevation is determined based on the depth difference between the completed well depth and the top of the Ordovician formation.**

Pressure potential refers to the potential energy of subsurface fluids resulting from their own pressure (Pang et al., 2007). When the variation in fluid density is negligible, the pressure potential at the production interval can be simplified to the hydrostatic pressure at the wellbore, which is equivalent to the bottom-hole flowing pressure.

$$\Phi_2 = \rho\int_0^p \frac{dp}{\rho(p)} \approx \rho\int_0^p \frac{dp}{\rho} \approx \int_0^p dp = P = P_{wf} \tag{2}$$

Based on the previously described fluid potential characterization and calculation methods, the primary fluid potential at each well location was first determined. The fluid potential field in the inter-well regions was then estimated through interpolation. On this basis, fluid potential distribution maps were generated for each well group and the overall study area (Liu et al., 2014).

## Bottom-Hole Pressure Conversion

The Hagedorn and Brown model (hereinafter referred to as the HB model) was employed to calculate the multiphase flow pressure drop along the wellbore. The bottom-hole flowing pressure was subsequently estimated from the wellhead pressure using the calculated pressure gradient, allowing for the determination of the pressure potential at each well location (Hagedorn and Brown, 1965). The pressure gradient equation of the HB model, as presented in Equation (3), includes three components: the hydrostatic pressure drop, the frictional pressure drop, and the acceleration pressure drop.

$$\begin{aligned}\left(\frac{dP}{dz}\right)_{\text{total}} &= \left(\frac{dP}{dz}\right)_{\text{hydrostatic}} + \left(\frac{dP}{dz}\right)_{\text{friction}} + \left(\frac{dP}{dz}\right)_{\text{acceleration}} \\ &= \rho_m g + f_m \frac{G_m^2}{2DA^2\rho_m} + \frac{d}{dz}\left(\frac{G_m^2}{\rho_m A^2}\right)\end{aligned} \tag{3}$$

$$\rho_m = \rho_L H_L + \rho_g\left(1 - H_L\right) \tag{4}$$

$$G_m = G_\text{g} + G_\text{L} = A\left(v_{SL} + v_{SG}\rho_g\right) \tag{5}$$

where $\rho_g$, $\rho_L$, $\rho_m$ represent the densities of gas, liquid, and the gas–liquid mixture, respectively, in kg/m$^3$. $H_L$ is the liquid holdup, and is the gravitational acceleration (m/s$^2$). is the cross-sectional area of the pipe flow $\left(\pi D^2/4\right)$, where $D$ is the inner diameter of the pipe (m). $G_m$ denotes the total mass flow rate of the gas–liquid mixture (kg/s), while $G_g$ and $G_L$ are the mass flow rates of gas and liquid, respectively (kg/s). $v_{SG}$ and $v_{SL}$ are the superficial velocities of gas and liquid (m/s), calculated as $v_{SG} = q_g / A$ and $v_{SL} = q_L \big/ A \cdot f_m$. $f_m$ is the two-phase friction factor, which is calculated using the Jain equation.

When calculating the pressure gradient, the key step is to determine the liquid holdup $H_L$. This process requires introducing four dimensionless parameters: the liquid velocity number (NLV), gas velocity number (NGV), liquid viscosity number (NL), and pipe diameter number (ND). The HB model provides a graphical method for determining the liquid holdup. First, the dimensionless parameters are calculated based on static and dynamic data. Then, the NL–CNL chart is used to obtain the CNL value by based on the calculated NL. Subsequently, the CNL value is substituted into Equation (6) to determine the ratio $H_L/\Psi$ from another chart. The correction factor Ψ is obtained from a separate chart based on the results of Equation (7). Finally, the values of $H_L/\Psi$ are substituted into Equation (8) to calculate the liquid holdup $H_L$.

$$\left(N_{LV}/N_{GV}^{0.575}\right)\left(\frac{P}{P_G}\right)^{0.10}\frac{CN_L}{N_D} \tag{6}$$

$$\left(N_{GV}N_L^{0.38}\right)/N_D^{2.14} \tag{7}$$

$$H_L = \left(H_L/\Psi\right) \bullet \Psi \tag{8}$$

After obtaining the liquid holdup $H_L$, it is substituted into the pressure gradient equation for further calculation. By applying the relationship between depth increment, pressure increment, and pressure gradient, iterative calculations are performed using Equation (9) to finally determine the pressure drop within the wellbore.

$$\Delta Z_i = \Delta p \bigg/ \left(\frac{dp}{dZ}\right)_i \tag{9}$$

Based on the above methodology, an iterative calculation was performed using segmented pipe length increments, and a dedicated computational module was developed accordingly. In total, nine bottom-hole pressure tests were conducted in Well S9 of the T10 reservoir. The bottom-hole pressures calculated using the Hagedorn–Brown model were compared with the field-measured data (see Fig. 2). The comparison results indicate that, except for a few outliers, the relative errors of all test points were within 5%, meeting the accuracy requirements for engineering applications. These results fully validate the accuracy and applicability of the pressure conversion method proposed in this study.

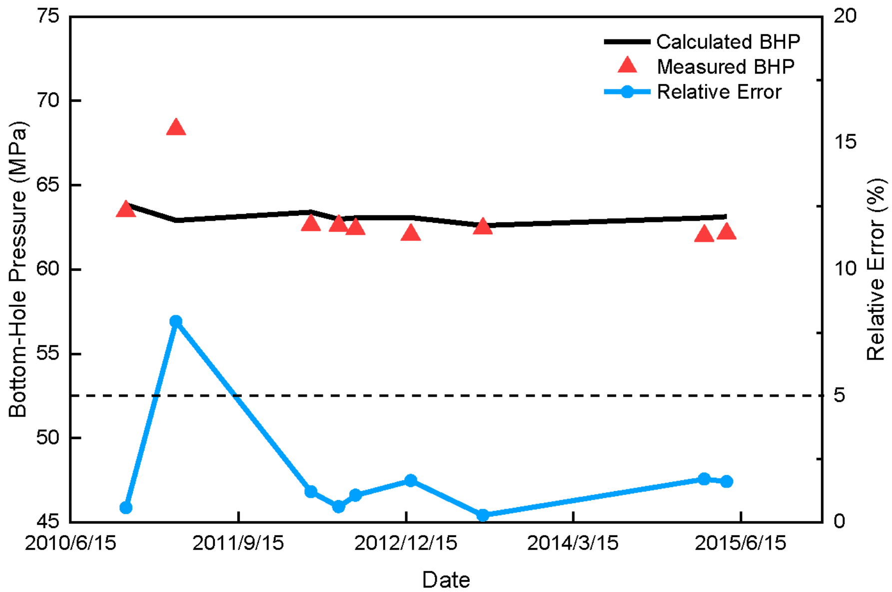


**Figure 2—The calculated bottom-hole pressures using the HB model are in good agreement with the measured field data, indicating that the model provides reliable pressure predictions.**

## Fluid Potential Analysis Workflow and Adjustment Strategy

The geological conditions of fractured-caved carbonate reservoirs are highly complex, with considerable variations not only between well groups but also among individual wells within the same group. Fluid flow and remaining oil distribution are influenced by a combination of factors that interact in a complicated manner. Therefore, prior to conducting fluid potential field analysis, it is necessary to systematically review and comprehensively integrate both static and dynamic reservoir data. This includes reservoir space types, water cut characteristics of production wells, inter-well connectivity, well-specific dynamic reserves, and aquifer multiples. Such comprehensive evaluation provides a solid foundation for formulating targeted and technically sound fluid potential adjustment strategies.

### Production Performance Analysis

The production performance analysis primarily focuses on identifying the reservoir space types and water production characteristics. By integrating static data interpretation, such as seismic profiles and well logging curves, with drilling fluid loss observations and production performance data, the reservoir space types and their influence on fluid flow can be comprehensively evaluated. Based on this analysis, the reservoir characteristics of individual wells in the target area were systematically classified. The reservoirs can be divided into two categories: cave-type reservoirs and fracture-vuggy reservoirs. Cave-type reservoirs are dominated by large karst caves and are typically associated with wellbore blowouts and significant drilling fluid losses during drilling. On seismic profiles, these reservoirs are characterized by the typical "beaded" reflection pattern (Fig. 3a). In contrast, fracture-vuggy reservoirs are primarily controlled by fracture networks partially connected to small karst caves. Drilling fluid losses in these reservoirs are minor or may not occur, and their seismic responses generally exhibit "sheet-like" reflection patterns (Fig. 3b).

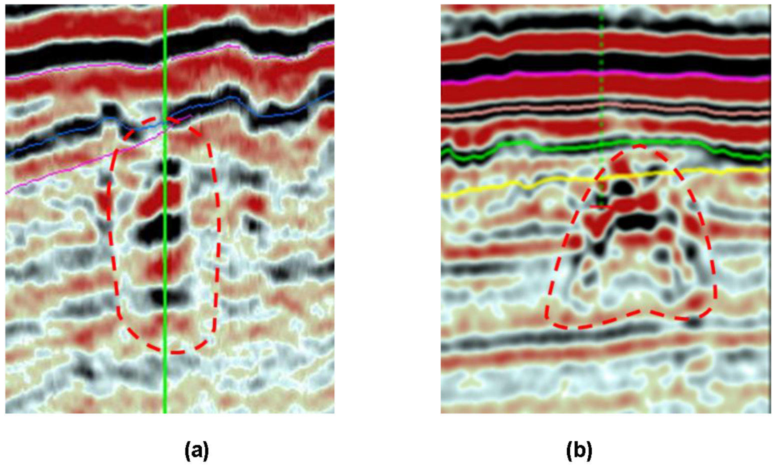

Figure 3—Seismic profiles of different reservoir types. (a) The cave-type reservoir is characterized by a typical "beaded" reflection pattern. (b) The fracture-vuggy reservoir generally exhibits a "sheet-like" reflection pattern.

Water production behavior can be effectively identified by analyzing water cut variation curves during the natural flowing stage of oil wells. This provides a key basis for evaluating the connectivity between fracture-cave units and aquifers, as well as assessing the strength of aquifer energy, which is essential for guiding injection–production adjustment strategies in the mid-to-late stages of reservoir development. Through systematic production performance analysis, five typical water cut variation patterns were identified in the study area: step-increase type, rapid water breakthrough type, intermittent water production type, fluctuation type, and no water production type. The step-increase type is characterized by a sudden rise in water cut after a period of stable production, indicating that bottom water has broken through and formed a stable water channel. The rapid water breakthrough type is identified by early water production during the initial development stage, with a rapid increase in water cut over a short period, suggesting strong aquifer energy and high oil–water connectivity. The intermittent water production type is typically reflected by alternating cycles of water production and no water production, often caused by poor connectivity within fracture-cave units or strong heterogeneity in seepage channels. The fluctuation type is marked by continuous variations in water cut, indicating dynamic changes in aquifer energy. The no water production type is characterized by a consistently low water cut over an extended period, implying that the fracture-cave units are either not connected to the bottom aquifer or are extremely poorly connected.

## Inter-Well Connectivity Evaluation

Inter-well connectivity evaluation is a critical foundation for constructing fluid potential fields, identifying fluid flow channels between fracture-cave units, and formulating effective adjustment strategies. To accurately quantify the degree of connectivity between wells, this study employs the INSIM model for connectivity analysis(Wang et al., 2019; Zhao et al., 2019). In this model, the reservoir layers between the target wells are discretized into a series of connected units, each characterized by inter-well transmissibility $T_{ij}$ and connected pore volume $V_{\mathrm{P,i,j}}$. Based on the principle of material balance, a material conservation equation is established for each water injection well, accounting for the connectivity with surrounding production wells. This equation serves as the foundation for subsequent connectivity evaluation and fluid distribution calculations.

$$\sum_{j=1}^{N_W} T_{ij}(t)\left[p_j(t)-p_j(t)\right]+q_i(t)=\frac{dp_i(t)}{dt}C_tV_i(t) \tag{10}$$

The material conservation equations were reorganized and discretized using the implicit finite difference method to establish a fundamental system of equations for solving the pressure field. During the calculation process, it is necessary to alternate between constant-rate and constant-pressure boundary conditions based on actual production scenarios. Under constant-rate conditions, the source/sink term $q_i^n$ is introduced into the system as a known constant, and the pressure field matrix equation is constructed by incorporating the transmissibility relationships between multiple connected wells. Solving this matrix yields the nodal pressure distribution across the connected units. Based on the calculated pressure gradients, the flow direction and flow rate within each connected unit can be further determined, as expressed by the following equation:

$$q_{ij}^n = T_{ij}^n\left(p_j^n - p_i^n\right) \tag{11}$$

Under constant-pressure boundary conditions, the transmissibility index in the injection–production direction between wells must first be determined. Based on this, the corresponding system of pressure field equations is constructed to calculate the average pressure within each connected unit. Once the pressure distribution is obtained, the injection and production rates for individual wells can be back-calculated using the following equation.

$$q_i^n = J_i^n\left(p_i^n - p_{wfi}^n\right) \tag{12}$$

To accurately represent the flow behavior within the connected units, a two-phase steady-state flow regime is assumed. The Buckley–Leverett frontal advance theory is applied to describe the displacement process and to calculate the flow boundaries within each unit. Based on the pressure field solution, the flow direction and upstream–downstream relationships between wells can be further clarified. By combining the water cut information from the upstream wells with the cumulative pore volume multipliers of the connected units, the water cut between Well $i$ and Well $j$ can be estimated using the following equation.

$$f'_w\left(S_{wij}^n\right) = f'_w\left(S_{wij}^n\right) + \frac{1}{F_{vij}^n} \tag{13}$$

After sequentially calculating the water cut between Well $i$ and each of its connected wells, the overall water cut for Well $i$ is determined by performing a flow-rate-weighted average, as expressed by the following equation:

$$f_{wi}^n = \frac{\sum_{j=1}^{N_{wn}} q_{ij}^n f_{wij}^n}{\sum_{j=1}^{N_{wn}} q_{ij}^n} \tag{14}$$

Ultimately, the flow allocation factors between injection and production wells can be determined through the connectivity analysis described above. By incorporating the current water cut and other dynamic reservoir data, the injection–production ratio and well group operating parameters can be further optimized and adjusted to improve the overall system performance.

## Dynamic Reserves and Aquifer Size Estimation

In this study, a quantitative method for evaluating dynamic reserves and aquifer size in water-driven reservoirs was applied to estimate the dynamic reserves and aquifer volume for individual wells (Li et al., 2017). Based on the extent of energy sweep and production performance characteristics, the production life of oil wells was divided into three stages: pre-water breakthrough, early water breakthrough, and mid-to-late water breakthrough. Considering the water invasion behavior of the target reservoir, four diagnostic plots were established: the Blasingame water invasion identification plot, the flowing material balance plot, the oil pressure drop versus cumulative fluid production plot, and the pseudo $N_{pr}$ diagnostic plot (Fig. 4).

By analyzing the production performance of individual wells and referencing these diagnostic plots, the production stage of each well was identified.

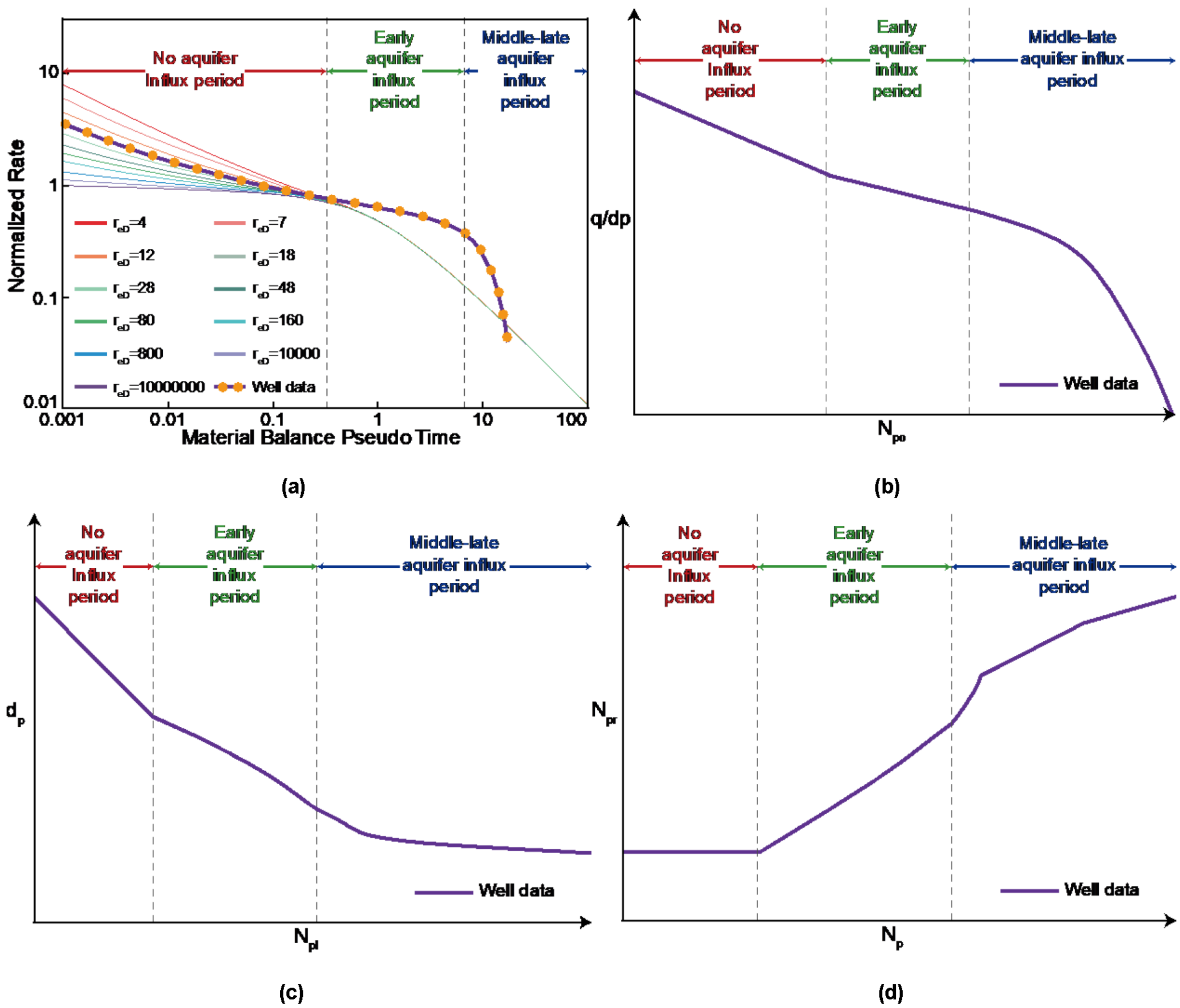


**Figure 4—Water invasion diagnostic plots for water-driven fractured-caved carbonate reservoirs. (a) Blasingame water invasion identification plot. (b) Flowing material balance water invasion identification plot. (c) Oil pressure drop versus cumulative fluid production plot for water invasion identification. (d) Pseudo Npr water invasion identification plot.**

The production data from the pre-water breakthrough stage were then fitted using a production decline analysis method that excludes the effect of water influx, in order to determine the dynamic geological reserves for each well. Subsequently, the water invasion index **J** was calculated using well-specific static and dynamic data to quantify the extent of water influx. Based on the fitted dynamic reserves and the calculated water invasion index, the production performance during the early water breakthrough stage was further matched using either the Blasingame method or the material balance method with aquifer influence, enabling the quantitative estimation of aquifer size.

## Fluid Potential Adjustment Strategies

Fluid flow in fractured-caved carbonate reservoirs is predominantly governed by the distribution of the fluid potential field. By adjusting the fluid potential profile, it is possible to effectively control bottom-water coning and water channeling, thereby enhancing the recovery of remaining oil and improving overall reservoir performance. Common fluid potential adjustment strategies include liquid lifting and drainage, liquid rate control for water coning suppression, and water flooding optimization. These strategies are selectively applied based on specific water invasion patterns and energy distributions within well groups.

The liquid lifting and drainage strategy, illustrated in Fig. 5a, is typically used for multi-well groups with strong bottom-water energy and high aquifer multiples. In such well groups, some production wells directly encounter karst caves connected to bottom-water, which leads to rapid water breakthrough and a sharp increase in water cut during the early production stage. As production continues, bottom-water expands along fracture networks driven by the fluid potential gradient, which limits the recovery of remaining oil in surrounding wells and increases the risk of water invasion. To mitigate this, increasing the liquid production rate in high-water-cut wells helps preferentially release bottom-water and lowers the pressure potential within the connected units, thereby achieving a pressure relief and drainage effect. This approach is proven to delay water cone development, restrict bottom-water expansion, stabilize the water cut in adjacent wells, and improve the overall recovery efficiency.

In Fig. 5b, the liquid rate control strategy for water coning suppression is presented. This method is generally applied to individual wells or well groups where the fracture-cave systems are interconnected primarily through fracture networks. These reservoirs typically exhibit large aquifer multiples and strong bottom-water energy. After water breakthrough, the water cut rises rapidly, and the well group usually enters a medium to high water cut production stage, while substantial remaining oil still exists around and between the wells. The core of this strategy involves reducing the liquid production rates to lower the pressure potential within the connected aquifer units, complemented by operational measures such as well shut-in or soaking to re-establish the initial oil–water balance. This technique effectively stabilizes water cones and improves oil recovery.

The water flooding optimization strategy, as depicted in Fig. 5c, is particularly suited for well groups operating under a one-injection multi-production pattern where production wells show uneven displacement responses and the sweep efficiency of injection wells is insufficient. In these cases, injected water tends to advance preferentially along dominant flow channels, leaving secondary flow paths or distant production wells poorly swept and limiting the overall development effectiveness. The success of this strategy depends on accurately identifying the location, scale, and connectivity of dominant and secondary flow channels between injection and production wells. By adjusting the injection–production scheme, modifying injection intensity, and optimizing the operational parameters of production wells, the injected water can be redirected towards under-swept secondary flow paths, thereby improving the performance of low-efficiency wells and enhancing the recovery of remaining oil.

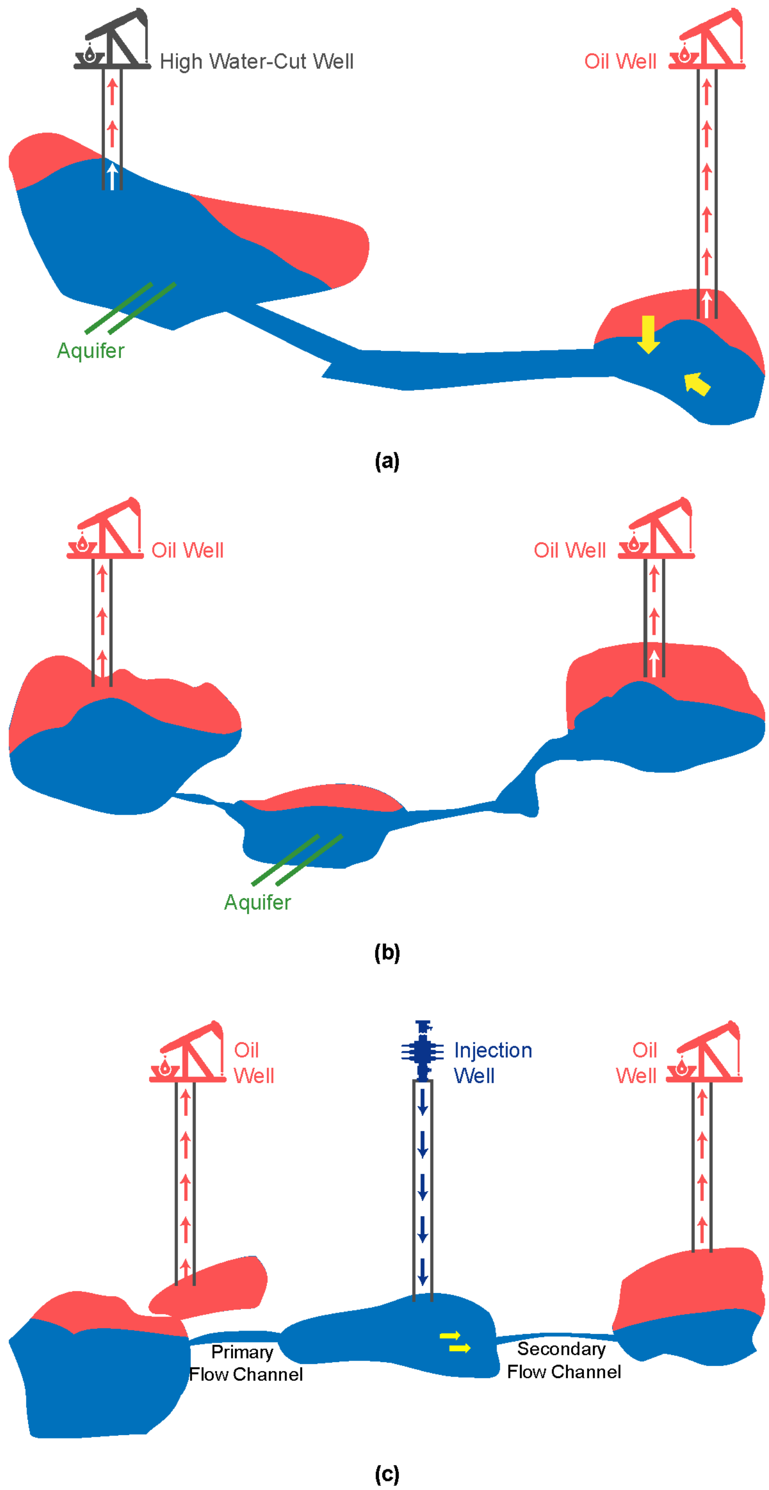


**Figure 5—Schematic diagrams of fluid potential adjustment strategies. (a) The liquid lifting and drainage method guides bottom-water outflow and reduces pressure potential. (b) The liquid rate control method stabilizes water cones by reducing production rates and restoring oil–water balance. (c) The water flooding optimization method redirects injected water to improve sweep efficiency and remaining oil recovery.**

## Case Study

### Analysis of a Typical Well Group

Based on the analysis methodology described above, the T10336 well group in the T10 oilfield of the Tarim Basin, China, was selected as the case study for fluid potential calculation and adjustment strategy evaluation. A comprehensive analysis of production performance data for individual wells within this group was performed. The results indicate that most of the ten wells encountered fracture–vuggy type reservoirs, while the others are primarily associated with cave-type reservoirs. Water production behavior in this group is predominantly characterized by intermittent water production and fluctuation patterns, although other water cut responses are also present, reflecting considerable variability across the wells. This suggests significant differences in the scale, spatial structure, and reservoir properties of the storage units encountered by individual wells, highlighting the strong heterogeneity and complex geological architecture of the reservoir. The connectivity evaluation results for the T10336 unit are shown in Fig. 6. In this figure, the color and thickness of the connection lines represent the transmissibility between wells. Seismic interpretation and image logging data further reveal that a high-angle fracture intersects Well TH12362CH and Well TH10343, indicating good connectivity and high transmissibility between the two wells. Another high-angle fracture was identified passing through Well TH10246CH and extending near Water Injection Well TH10383, suggesting that TH10383 exhibits the strongest connectivity with TH10246CH. In addition, TH10383 shows good connectivity with the northern production wells, but poor connectivity with Well TH10363 in the southern area, indicating that the northern section is likely to contain a greater number of secondary fractures.

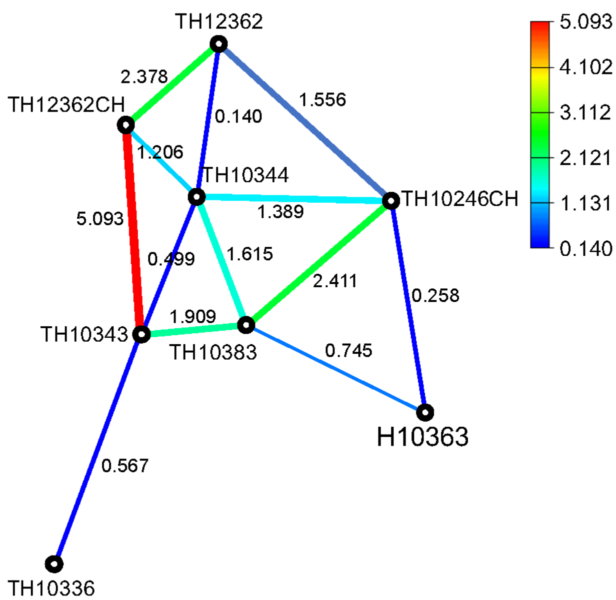


**Figure 6—Connectivity relationships and transmissibility distribution in the T10336 unit.**

The dynamic reserves, recovery factors, and aquifer multiples for individual wells in the T10336 unit were further evaluated and are presented in Fig. 7. The analysis reveals considerable variability in these parameters across the wells, reflecting strong reservoir heterogeneity, complex fluid migration pathways, and significant challenges for reservoir management in this area. Among the wells, Well TH10331CH holds the highest dynamic reserves, approximately $2.0 \times 10^6$ $m^3$, yet its recovery factor remains relatively low, indicating a substantial volume of remaining reserves and considerable potential for further development. Moreover, the low aquifer multiple suggests limited natural energy support from the formation. Given the well's position within the fluid potential field, water injection or water flooding strategies are recommended to enhance recovery efficiency. By contrast, Well TH10353H is associated with a relatively high aquifer multiple, lower dynamic reserves, and a moderate recovery factor. Considering its production profile and inter-well connectivity, this well is a candidate for potential conversion to a water injection well.

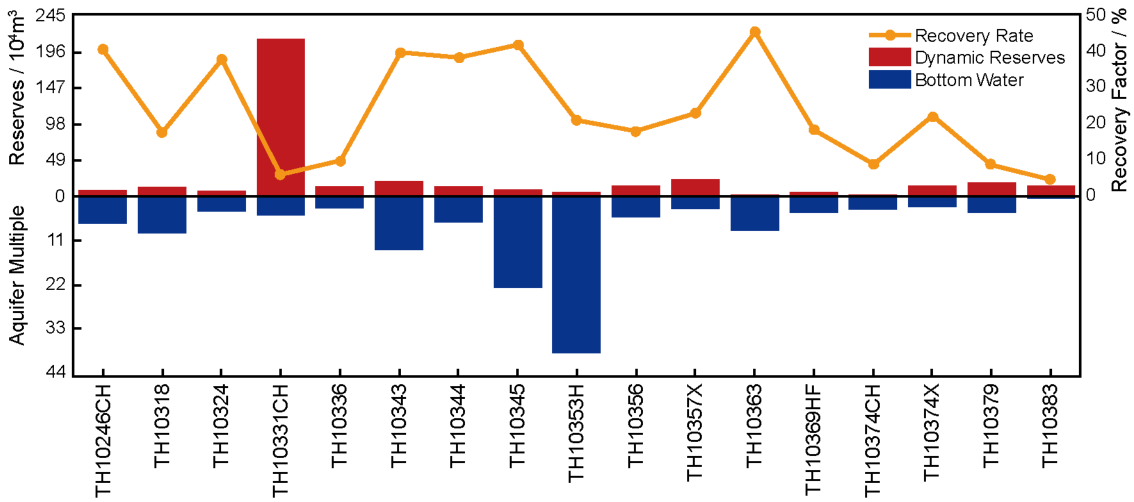


**Figure 7—Comparison of dynamic reserves, cumulative oil production, and aquifer multiples for wells in the T10336 unit.**

Based on the previous analysis and the fluid potential field distribution across the study area (Fig. 8), a differentiated fluid potential adjustment strategy was formulated. For Well TH10246CH, the current water cut is 60.97%, with a relatively low aquifer multiple of 4.62 and dynamic reserves of 78930 $m^3$. This well is situated in a low-potential zone and exhibits clear connectivity with a nearby water injection well, providing favorable conditions for sustained energy support. Considering the abundant remaining reserves, this well is recommended to continue as a primary production well, but it should be operated under a controlled liquid production strategy to effectively suppress water coning and delay its advancement. In contrast, Well TH10353H presents a significantly higher water cut of 99.15%, with dynamic reserves of only 56270 $m^3$ and a high aquifer multiple of 38.87. This well is characterized by fluctuating water cut behavior and is located within a medium-to-high potential zone. Due to the strong bottom-water energy and limited remaining oil potential, this well is more suitable for conversion into a drainage well, where increasing liquid production intensity can create a "pressure relief channel." This strategy is expected to stabilize water cones in surrounding wells and indirectly improve their production performance. Following the same approach, all wells within the T10336 well group were systematically evaluated and classified. The results indicate that four production wells should be operated under liquid rate control strategies to suppress water coning, while five wells are suitable for drainage and pressure relief operations. The remaining wells are recommended for adjustment through water flooding optimization to enhance sweep efficiency and further improve oil recovery.

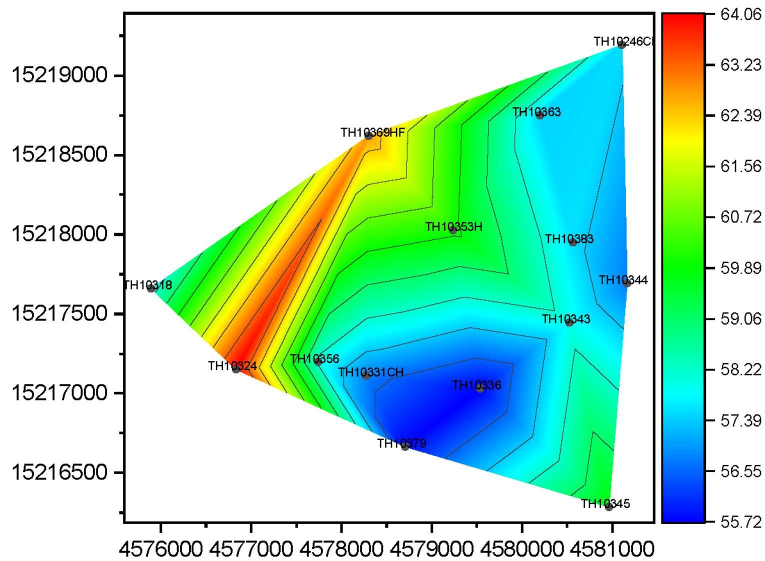


**Figure 8—Fluid potential field distribution for the T10336 well group.**

## Field-Wide Application

Based on the methodology described above, fluid potential calculations and adjustment strategy analyses were systematically carried out for all well groups across the study area. In total, 27 well groups and 158 production wells were evaluated. The analysis results indicate that 31 wells require liquid rate control to suppress water coning, 32 wells are suitable for drainage and pressure relief operations, and 34 wells should be adjusted using water flooding strategies. The evaluation also involved 25 water injection wells. It was observed that different adjustment strategies are controlled by different dominant factors, and the key decision-making parameters vary accordingly across strategies. To account for this variability, radar plots were constructed for each well group using the primary control parameters associated with the three adjustment strategies. Fig. 9 illustrates the radar plot for the liquid rate control strategy applied to the TH10303 well group, where the current water cut and aquifer multiple are the main influencing factors. A well distribution plot was generated using the current water cut, aquifer multiple, and adjustment strategy type as axes. The analysis shows that most wells already operating under the liquid rate control strategy are located along the outer edges of the radar plot, whereas wells not currently adjusted under this strategy are concentrated near the center. By synthesizing key parameter thresholds, a boundary line was established. Wells with parameters simultaneously exceeding this threshold can be identified as suitable candidates for implementing the liquid rate control strategy.

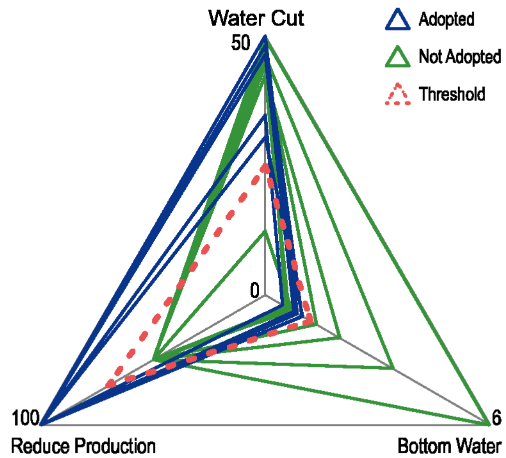


**Figure 9—Radar plot of the liquid rate control strategy for water coning suppression in the TH10303 well group.**

Based on the analysis results, the threshold boundaries for the three adjustment strategies were summarized for each well group, and a fluid potential adjustment policy chart was ultimately developed for the entire field (Fig. 10). This chart provides a practical reference for quickly determining whether an individual well requires the implementation of a specific adjustment strategy.

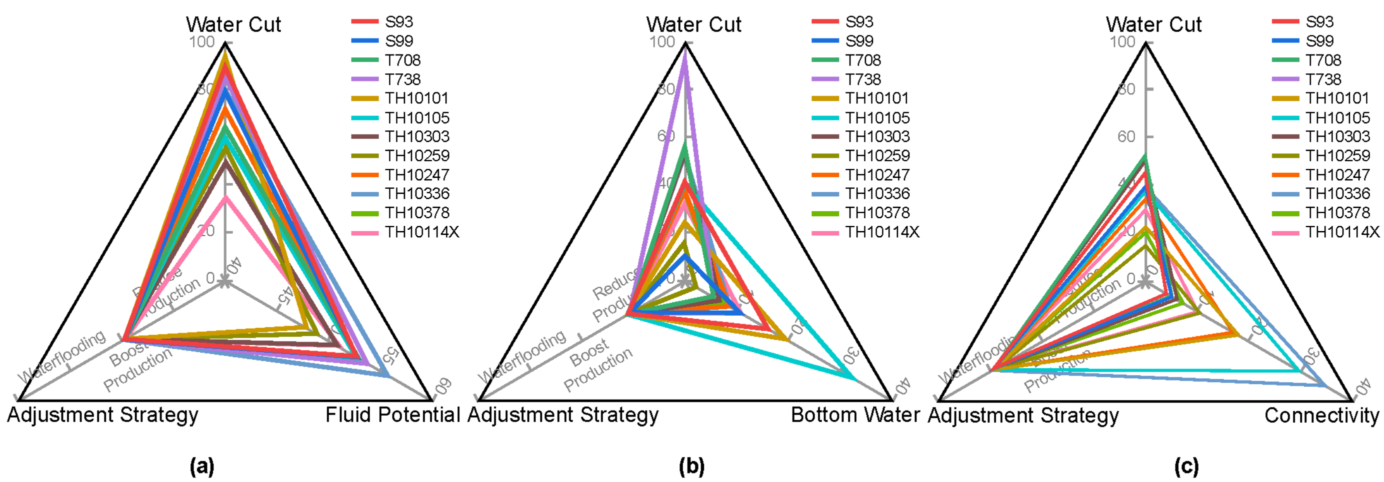


**Figure 10—Fluid potential adjustment policy chart for the entire T10 oilfield. (a) Liquid lifting and drainage strategy. (b) Liquid rate control strategy for water coning suppression. (c) Water flooding adjustment strategy.**

The adjustment strategies identified in this study were progressively implemented across 27 well groups in the field. Production data indicate that all adjustment strategies achieved effective water control and oil production improvement. In well groups where the liquid rate control strategy for water coning suppression was applied, the average water cut decreased by 6.3%, and daily oil production increased by 8.1%. For well groups where the liquid lifting and drainage strategy was adopted, increasing the liquid production rates of high-water-cut wells effectively guided the preferential withdrawal of bottom water, reduced the pressure potential within the connected flow units, and successfully curbed the rising water cut trend in nearby wells. This adjustment led to a 6.7% increase in daily oil production. In well groups where the water flooding adjustment strategy was implemented, optimizing injection intensity redirected injected water toward previously under-swept low-efficiency zones, expanded the sweep area, and resulted in a 7.4% increase in daily oil production. Between 2020 and 2024, the cumulative oil increment achieved across the entire study area reached approximately 746000 t, while the average water cut was consistently maintained within the range of 75% to 85%. These field results comprehensively demonstrate the practical effectiveness and wide applicability of the fluid potential evaluation and adjustment methodology proposed in this study.

## Conclusions

This study addresses key development challenges in fractured-caved carbonate reservoirs, including strong reservoir heterogeneity, complex fluid flow mechanisms, and significant water control difficulties. A multi-parameter coupled fluid potential evaluation and adjustment method was developed based on hydrodynamic theory and successfully applied in the T10 oilfield of the Tarim Basin, China. The main conclusions are as follows:

1. Fluid potential in fractured-caved carbonate reservoirs is primarily controlled by both gravitational potential and pressure potential. The accuracy of pressure potential evaluation is highly dependent on the precise calculation of bottom-hole flowing pressure. In this study, the Hagedorn & Brown model was introduced to calculate multiphase flow pressure losses in the wellbore, which significantly improved the accuracy and reliability of fluid potential evaluation.
2. The fluid potential field distribution was calculated based on a comprehensive analysis of reservoir space types, water cut patterns, inter-well connectivity, dynamic reserves, and aquifer multiples. The spatial characteristics of the fluid potential field were clearly defined at the well group level. Flow channel hierarchies were also classified, providing a solid foundation for the development of targeted adjustment strategies.
3. Three fluid potential adjustment strategies were systematically established: liquid lifting and drainage, liquid rate control for water coning suppression, and water flooding optimization. The applicability and parameter thresholds for each strategy were clearly defined. A radar plot-based rapid screening method was proposed, and a field-wide fluid potential adjustment policy chart was developed. This enabled the visual and efficient application of adjustment strategies across the reservoir.
4. The proposed adjustment strategies were progressively implemented across 27 fractured-caved units in the T10 oilfield. Field results demonstrate that, in well groups where the liquid rate control strategy was applied, the average water cut decreased by 6.3% and daily oil production increased by 8.1%. The liquid lifting and drainage strategy and the water flooding optimization strategy resulted in daily oil production increases of 6.7% and 7.4%, respectively. These results confirm that all three strategies effectively improved water control and oil production. From 2020 to 2024, the cumulative incremental oil production in the study area reached approximately 746000 t. The average water cut was consistently maintained between 75% and 85%. These outcomes fully validate the adaptability and field applicability of the proposed method.

The fluid potential calculation and adjustment framework established in this study provides practical technical support for water control, production stabilization, and differentiated management in fractured-caved carbonate reservoirs. This approach offers valuable guidance for the development optimization of similar complex reservoirs.

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